Contact Manager Backend – Project Documentation

Project Title: Contact Manager Backend Application

Team Lead:

Faikha Thabasum. S

Email: s.faikhathabasum@gmail.com

Team Members:

Sushvamitha. Y

Email: sushmasaran672@gmail.com

Bairavi. S

Email: bairavis765@gmail.com

Jeevitha. M

Email: mjeevitha0105@gmail.com

Project Description

The Contact Manager Backend is a web application developed using Node.js and Express.js that allows users to manage contact information efficiently. The system enables users to create, view, update, and delete contact details. The backend interacts with a MongoDB database to store and retrieve contact data.

This project demonstrates how RESTful APIs are developed and connected to a database using modern backend technologies.

Objectives

- To develop a backend system for managing contacts.
- To implement CRUD operations using REST APIs.
- To connect a Node.js application with MongoDB.
- To organize backend code using MVC architecture.

Technologies Used

- **Node.js** – JavaScript runtime environment
- **Express.js** – Web framework for Node.js
- **MongoDB** – NoSQL database
- **Mongoose** – ODM for MongoDB
- **CORS** – Cross-origin resource sharing middleware
- **Nodemon** – Automatically restarts server during development

Project Structure

The project follows a modular structure to organize the backend components.

```
contactmanager
└── server
    ├── src
    │   ├── controllers
    │   ├── dbConfig
    │   ├── middleware
    │   ├── models
    │   ├── routes
    │   ├── utility
    │   └── server.js
    ├── package.json
    └── package-lock.json
```

Explanation of Folders

controllers
Contains functions that handle business logic and process API requests.

dbConfig
Responsible for connecting the application to the MongoDB database.

middleware
Contains middleware functions such as authentication or error handling.

models
Defines database schemas using Mongoose.

routes
Contains API routes that connect HTTP requests to controller functions.

utility
Includes helper functions used across the application.

server.js
Main entry point of the backend application that starts the Express server.

Features

- Add new contact
- View contact list
- Update contact details
- Delete contact
- REST API based architecture
- MongoDB database integration

API Operations

Create Contact

Adds a new contact to the database.

Read Contacts

Retrieves all stored contacts from the database.

Update Contact

Allows modification of existing contact information.

Delete Contact

Removes a contact from the database.

How to Run the Project

Step 1: Install Node.js

Download and install Node.js from the official website.

Step 2: Install Dependencies

Navigate to the server folder and run:

npm install

Step 3: Start the Server

Run the following command:

npm start

Step 4: Access the Application

The server will start on:

http://localhost:5000

Advantages

- Simple and scalable backend architecture
- Efficient data management using MongoDB
- RESTful API design
- Easy integration with frontend applications

Conclusion

The Contact Manager Backend project demonstrates the implementation of a RESTful API using Node.js, Express, and MongoDB. It provides an efficient way to manage contact information while following best practices in backend development.